

Sumit Kumar

Data Engineer | NIT Patna '21

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Skills

Languages: Python, Java, Scala

Data: Spark, PySpark, SQL, Kafka, Kafka Connect, Debezium CDC, Airflow

Cloud: AWS (ECS, MSK, DynamoDB, DocumentDB, S3, Bedrock), GCP (BigQuery, Dataproc, Dataflow)

GenAI: OpenAI GPT, AWS Bedrock, LangChain, Prompt Engineering

Frameworks: Spring Boot, FastAPI, React

Tools: Git, Docker, Jenkins, Maven

Experience

Cocoblu

Feb 2026 – Present

Senior Manager – DSIT (Data Science & IT)

Bangalore, India

- Engineered a product classification pipeline utilizing **Gemini 2.5 Flash** and **AWS Bedrock** to process **50,000+ ASINs**. Implemented global image deduplication and fuzzy-matching algorithms, reducing LLM token costs by **93%** and increasing classification accuracy to **96.6%**.
- Architected a Master Data Management (MDM) platform using **React** and **FastAPI** to unify organizational data workflows. Engineered a centralized ingestion backend utilizing **AWS Redshift** staging tables and **S3 COPY**, enabling high-volume asynchronous bulk upserts and deprecating isolated legacy data pipelines.
- Optimized organizational data pipelines utilizing **AWS Step Functions** to eliminate redundant execution steps. Engineered an **AI agent-based** pricing pipeline that performs competitor product analysis and interfaces with **Amazon SP-API** to execute continuous, automated price adjustments across the product catalog.
- Architected a centralized **Self-Healing AI Agent** and enterprise automation SDK using **AWS Lambda** and **Amazon Bedrock (Claude)**. Engineered an autonomous CI/CD remediation loop that intercepts pipeline crashes, generates LLM-driven code fixes, executes test validations via **AWS CodeBuild**, and automatically submits Pull Requests.
- Engineered hybrid **Machine Learning and Operations Research (OR)** backend systems utilizing **PuLP (MILP)** and **ADMM engines** for supply chain automation. Implemented mathematical solvers and **EOQ models** to optimize purchasing decisions for continuous in-stock availability and automate ASIN deal shortlisting against business constraints.
- Engineered a Python SDK integrating **SP-API** domain interfaces to facilitate high-throughput data extraction and automated pricing pipelines. Built the networking layer with **SigV4 signing**, **token caching**, exponential backoff, and circuit breakers to handle API rate limits and transient failures.

Gen Digital (formerly NortonLifeLock)

Dec 2024 – Jan 2026

Data Engineer

Chennai, India

- Engineered custom **Java-based Kafka Connect SMT library** for Debezium CDC processing from AWS DocumentDB, achieving **100% data integrity** and eliminating critical DebeziumException errors across real-time document pipelines.
- Implemented multi-field validation and smart filtering in **Spring Boot** transaction normalization services, **reducing invalid data processing by 40%** and supporting Gen's Unified Data Model initiative on **AWS ECS/MSK**.
- Designed and deployed **Python event-driven microservice** on ECS for member data operations, integrating Kafka for event consumption, DynamoDB for persistence, with comprehensive **GDPR compliance** and GUID sync strategy.
- Architected **Financial Insights Provider** — a Python/FastAPI microservice with **DynamoDB caching (3x speedup)**, 6 insight generators (Fee Monitoring, Duplicate Detection), and intelligent merge-on-PATCH logic for cross-scan deduplication.

FINN

- Led development of a **full-stack module** within the financial wellness platform **leveraging OpenAI GPT** for behavioral analysis, proprietary emotion scoring intelligence, and **Monte Carlo simulations** for retirement planning.

PayPal (via TCS)

Jul 2021 – Nov 2024

Data Engineer

Bangalore, India

Data Migration Framework (Mar 2023 – Nov 2024)

- Developed scalable **ETL framework** for PayPal's data migration using Python, AWS, and BigQuery — achieved **20% faster migration** and **30% improved scalability**.

- Built automated dashboards with **Matplotlib** for stakeholder visibility and deployed orchestration using **Airflow** with auto-generated DAG scripts.

Lynx Entity Linkage Optimization *(Jan 2024 – May 2024)*

- Optimized **Lynx entity linkage framework** using Scala/Spark and Google's APSS algorithm — achieved **35% better accuracy** and **40% faster** approximate nearest neighbor search via LSH enhancements.
- Conducted comprehensive testing with ML algorithms (RPDBSCAN, K-Means) for similarity scoring validation.

On-Demand Merchant Reporting *(Aug 2021 – Jan 2023)*

- Built **on-demand merchant reporting system** with Python, Oracle DB, and GCP Dataproc — achieved **15% improved accuracy** and **25% faster** report generation.
- Created end-to-end pipeline integrating report requests, authentication, and automated orchestration via internal Airflow (DALM).

Education

National Institute of Technology Patna

Bachelor of Technology in Computer Science & Engineering

2017 – 2021

CGPA: 8.0/10